

High-Dimensional Iterative D-optimal Experiment for NN-FullPR Transfer

Paper-scale bilingual report update with n=10000, p=20/50, CPU/GPU, and multi-seed replication

Dataset scale	Original n=10000; training/evaluation split=7500/2500
Dimensions and cases	p=20,50; quadratic / smooth / strong nonlinear
Replication	3 data seeds × 2 NN initialization seeds × 2 initial fractions × CPU/GPU
Workload	72 NN oracle trainings; 144 iterative paths; 8640 random surrogate fits

Executive Summary

Main finding: At the final iteration, D-optimal upgrading outperformed the same-budget random-upgrade median in 128 of 144 replications. The final positive-rate was 88.9%, and the average MSE gain was 2.61%. The result supports iterative D-optimal design as a stable data-upgrading strategy for high-dimensional NN-to-FullPR surrogate construction, while also showing that the gain depends on feature expressiveness, initial sampling fraction, and nonlinearity strength.

Experimental Workflow

Generate a high-dimensional original dataset and split it into a training pool and an evaluation set.

Train an NN oracle on the training pool; this NN is treated as the new target model.

Uniformly sample the initial design set and estimate the pilot distance between the NN and the FullPR surrogate.

At each iteration, compute D-optimal leverage conditional on the already upgraded design set and add a new batch of samples.

Refit the FullPR surrogate after each upgrade and compare it against random upgrades under the same budget.

Experimental Matrix and Workload

Item	Setting
Original dataset size	10000
Input dimensions	20, 50
Synthetic cases	quadratic, smooth, strong nonlinear
Initial uniform fractions	0.05, 0.1
Iterations	5; 500 samples added per iteration
Random baseline	10 repeats per iteration point
Devices	CPU and NVIDIA GeForce RTX 4070 Ti SUPER
FullPR features	degree=2, include_special=True

Main Results

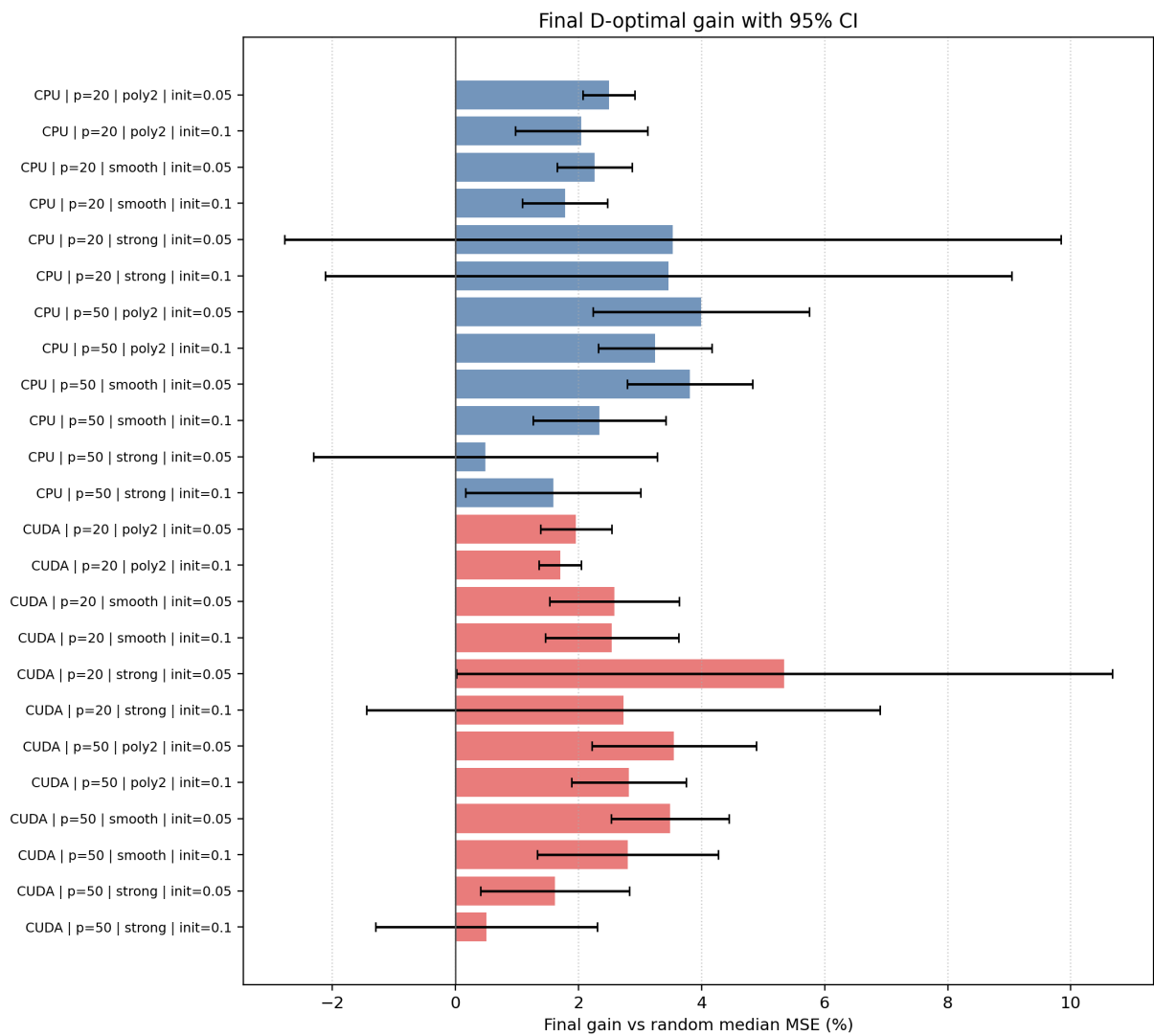


Figure 1. Final-iteration D-optimal gain over random upgrading, mean +/- 95% CI.

Final-Iteration Group Statistics

Tables 1 and 2 summarize final-iteration performance. A positive gain means that the D-optimal surrogate has lower MSE than the same-budget random-upgrade median.

Case	p	Init	Reps	Gain mean +/-95%CI	Positive
poly2	20	0.05	6	2.49% +/- 0.43%	100.0%
poly2	20	0.1	6	2.04% +/- 1.08%	83.3%
smooth	20	0.05	6	2.26% +/- 0.61%	100.0%
smooth	20	0.1	6	1.78% +/- 0.69%	100.0%

Case	p	Init	Reps	Gain mean +/-95%CI	Positive
strong	20	0.05	6	3.53% +/- 6.31%	66.7%
strong	20	0.1	6	3.46% +/- 5.58%	66.7%
poly2	50	0.05	6	3.99% +/- 1.76%	100.0%
poly2	50	0.1	6	3.24% +/- 0.92%	100.0%
smooth	50	0.05	6	3.81% +/- 1.02%	100.0%
smooth	50	0.1	6	2.34% +/- 1.08%	100.0%
strong	50	0.05	6	0.48% +/- 2.79%	66.7%
strong	50	0.1	6	1.59% +/- 1.42%	66.7%

Case	p	Init	Reps	Gain mean +/-95%CI	Positive
poly2	20	0.05	6	1.96% +/- 0.58%	100.0%
poly2	20	0.1	6	1.70% +/- 0.34%	100.0%
smooth	20	0.05	6	2.58% +/- 1.05%	100.0%
smooth	20	0.1	6	2.54% +/- 1.08%	100.0%
strong	20	0.05	6	5.35% +/- 5.33%	83.3%
strong	20	0.1	6	2.73% +/- 4.17%	66.7%
poly2	50	0.05	6	3.55% +/- 1.34%	100.0%
poly2	50	0.1	6	2.82% +/- 0.93%	100.0%
smooth	50	0.05	6	3.49% +/- 0.96%	100.0%
smooth	50	0.1	6	2.80% +/- 1.47%	83.3%
strong	50	0.05	6	1.62% +/- 1.21%	83.3%
strong	50	0.1	6	0.50% +/- 1.80%	66.7%

Iteration Curves

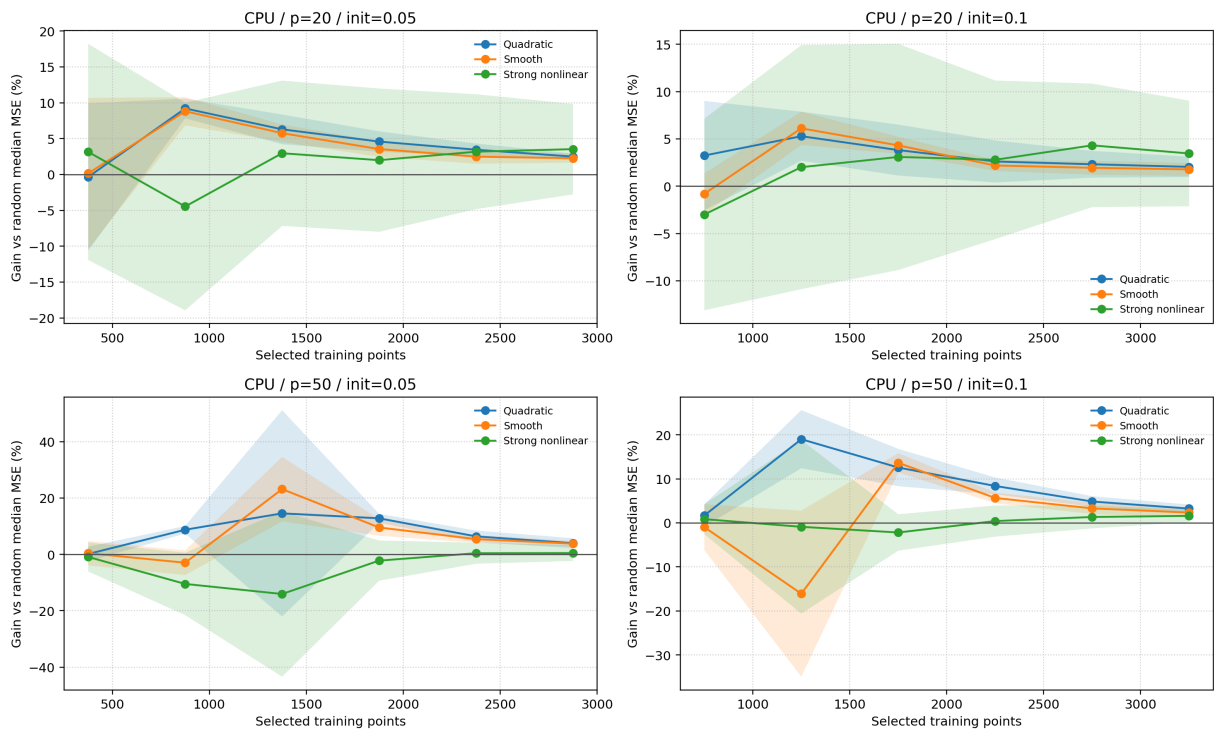


Figure 2. CPU gain curves with 95% confidence intervals.

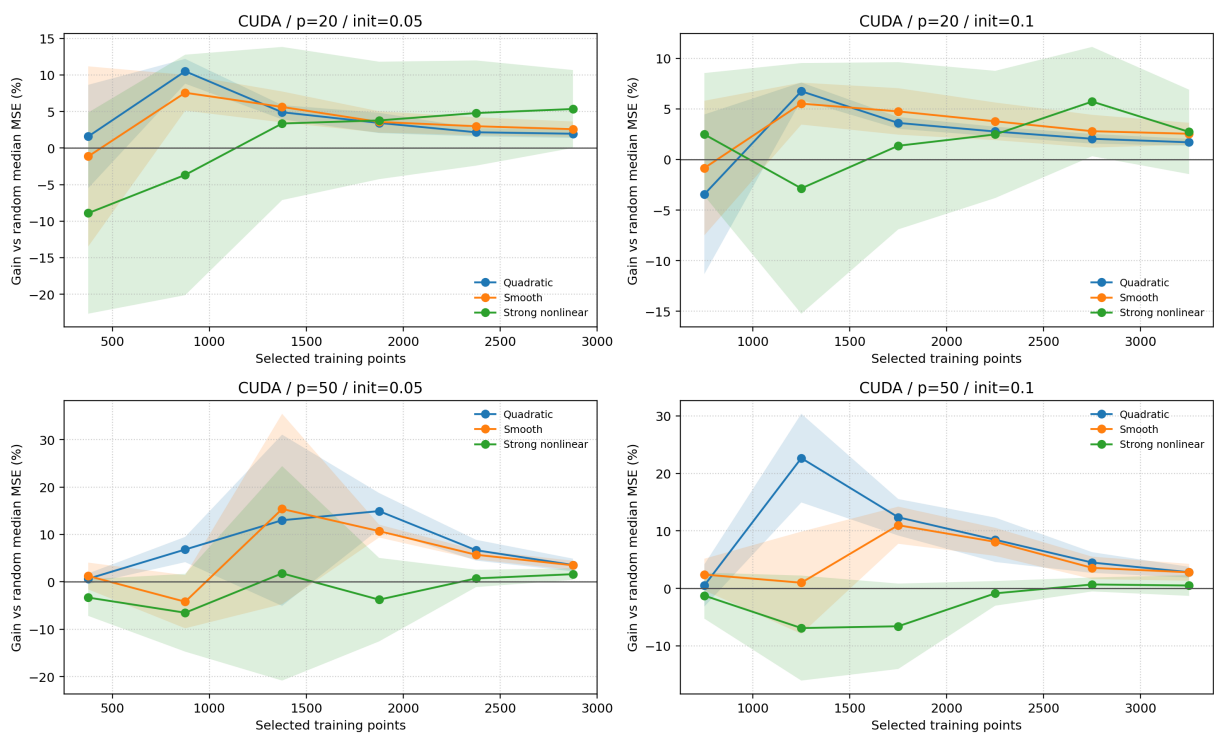


Figure 3. CUDA gain curves with 95% confidence intervals.

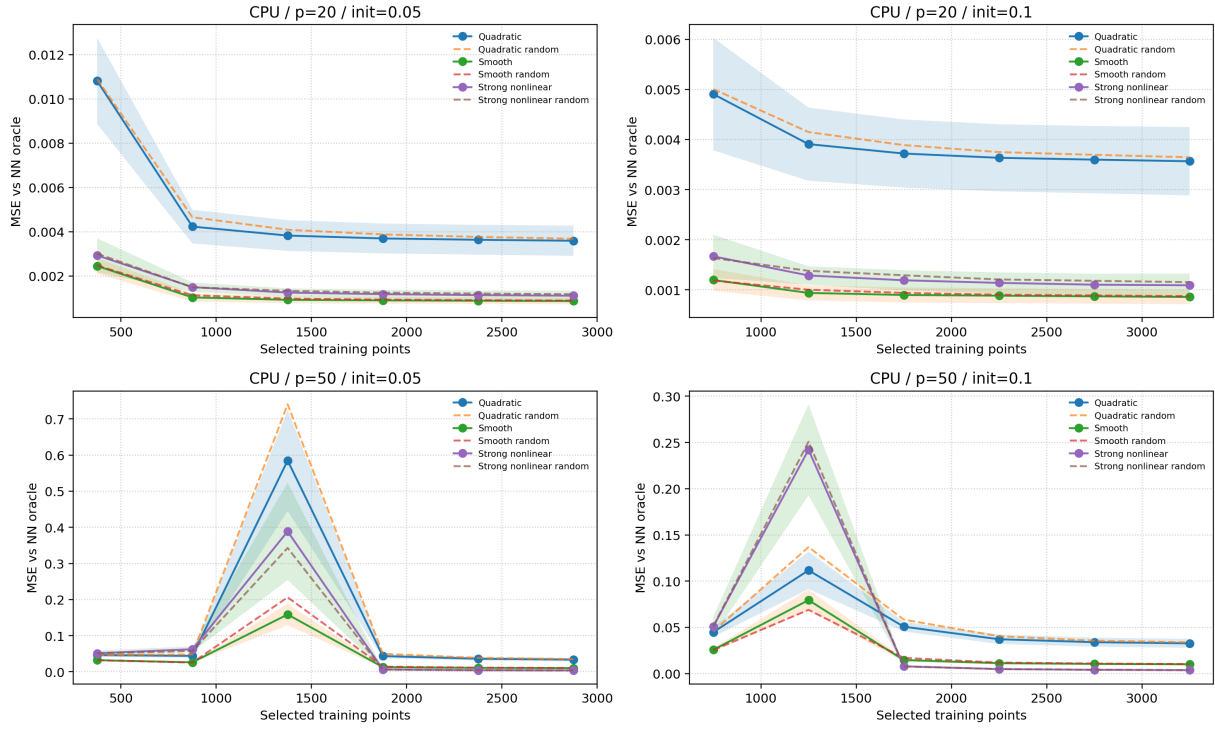


Figure 4. CPU MSE curves for D-optimal surrogates and random baselines.

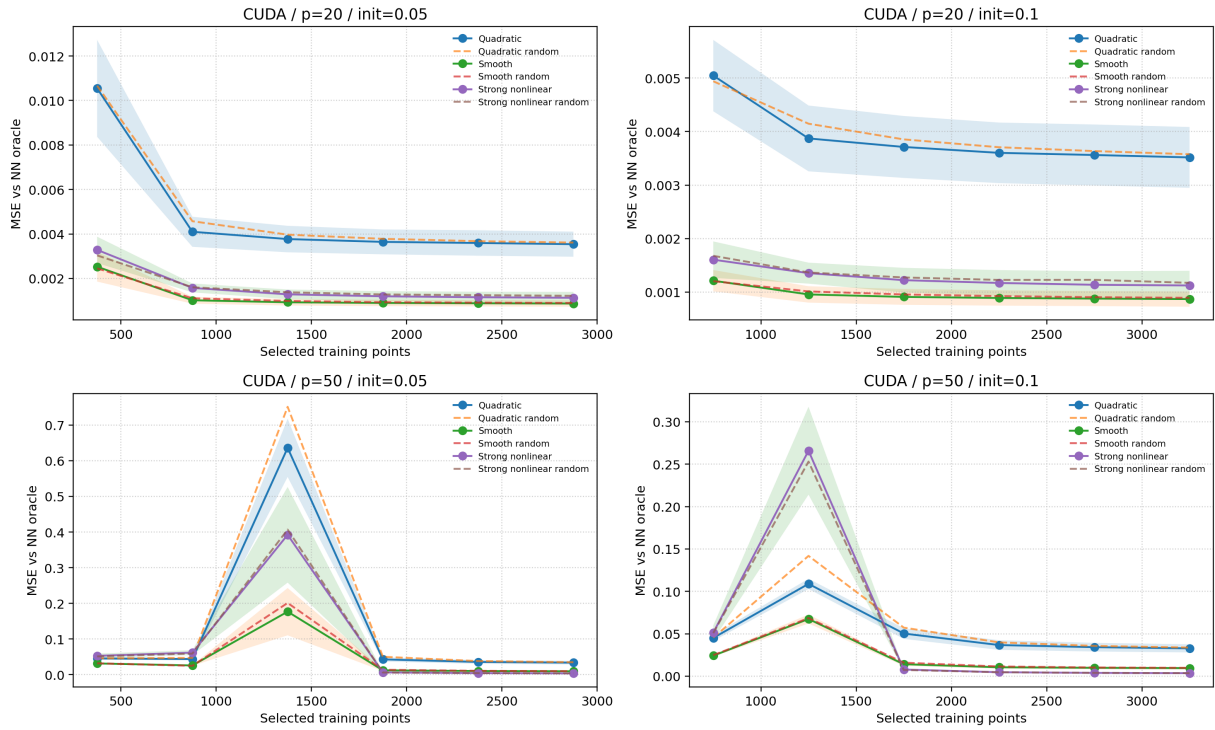


Figure 5. CUDA MSE curves for D-optimal surrogates and random baselines.

Discussion

For $p=20$, the quadratic FullPR feature count is relatively small, so the initial design often already exceeds the number of features. D-optimal gains are therefore modest but stable.

For $p=50$, the initial design is more likely to be under the FullPR feature count. Early iterations are more volatile, but the upgraded design stabilizes as it is reused in later D-optimal steps.

The strong nonlinear case has wider confidence intervals, which marks the boundary of the method: D-optimal selection can improve coverage, but it is not a guarantee that a low-degree FullPR surrogate fully captures the NN oracle.

CPU and GPU runs lead to the same qualitative conclusion, reducing the chance that the result is a device-specific artifact.

Conclusion

This updated experiment is suitable as a main paper experiment. The most defensible claim is that pilot distance and iterative D-optimal upgrading form an observable and reproducible diagnostic workflow for data selection. In high-dimensional settings, the workflow usually improves over random upgrading, but the magnitude of improvement depends on the FullPR feature space, nonlinearity strength, initial sampling fraction, and iteration budget.